Jiayi Chen

+86 18701381075 | jiayichen5@link.cuhk.edu.cn | Shenzhen, China
Google Scholar | LinkedIn | GitHub | Homepage | ORCID

EDUCATION

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) Sept. 2024 – 2029 (expected)
Ph.D. in Computer and Information Engineering, School of Science and Engineering (SSE) Shenzhen, China

- **Advisor**: Prof. Guangxu Zhu; **Co-advisor**: Prof. Shuai Wang

Beijing University of Posts and Telecommunications (BUPT) Sept. 2020 – Jun. 2024
B.Eng. in Internet of Things Engineering, Intl. School; QMUL-BUPT Joint Programme Beijing, China

- **Weighted Average Mark**: 90.07/100; **GPA**: 3.73/4.0; ranked **6/189**
- Selected into the YePeida Institute Pilot Class, admitting approximately the top **7%** of students

SELECTED PUBLICATIONS

Jiayi Chen et al. "WhichBench and WhichGround: Benchmarking and Solving 'Which One?' Problem in Exact-Instance Grounding for Embodied Agents." Submitted to *IEEE Robotics and Automation Letters (RA-L)*. [Video]

Jiayi Chen, Shuai Wang, Guangxu Zhu, and Chengzhong Xu. "Bridging Large-Model Reasoning and Real-Time Control via Agentic Fast–Slow Planning." *ICRA 2026*, pp. 13398–13405. [DOI] [Paper] [Code] [Video]

Jiayi Chen, Shuai Wang, Guoliang Li, Wei Xu, Guangxu Zhu, Derrick Wing Kwan Ng, and Chengzhong Xu. "Opportunistic Collaborative Planning with Large Vision Model Guided Control and Joint Query-Service Optimization." *IROS 2025*, pp. 951–958. [Paper] [Video]

Jiayi Chen, Shuai Wang, Guangxu Zhu, Derrick Wing Kwan Ng, Chengzhong Xu, and Kaibin Huang. "Not All Uncertainty Matters: Simulation-in-the-Loop Fast–Slow Reasoning for Decision-Critical Autonomous Driving System." *IEEE Transactions on Mobile Computing*, major revision. [Paper] [Code]

Jiayi Chen, Zhuo Li, and Guangxu Zhu. "RULER: Reasoning from Trials to Transferable Rules for Finite-Budget Online Algorithm Configuration." *ICLR 2027*, under review.

Jiayi Chen, Bin Wang, Jun Fang, Guangxu Zhu, and Qingjiang Shi. "Learning to Optimize Across Problem Scales: Size-Agnostic Solver Selection for Accelerated Constrained Optimization." *ICLR 2027*, under review.

Jiayi Chen, Weiting Ou, and Guangxu Zhu. "The Universal Language of CSI: Unifying Wireless Sensing Across Devices and Environments." *IEEE Communications Magazine*, 2026. [DOI] [Paper] [Code]

RESEARCH EXPERIENCE

Embodied Intelligence and Embodied Agents

- **Embodied Agent Grounding** May 2026 – Oct. 2026 *IEEE RA-L, Submitted*
  During the AgiBot–AGIQUAD internship, developed WhichBench to diagnose VLN capability gaps and inform a **targeted data feedback loop**. Built WhichGround as a **prototype embodied inspection agent**, orchestrating semantic-map/VLN systems as callable skills.
- **Natural Language to Low-Level Motion Planning** Mar. 2025 – Sept. 2025 *ICRA 2026, DOI*
  Developed AFSP to translate **natural-language instructions** into **dynamically feasible trajectories**, preserving semantic intent and improving robustness in real-time robot control.
- **Adaptive Fast–Slow Thinking** Dec. 2023 – Jun. 2026 *IROS 2025; IEEE TMC, Major Revision*
  Developed OCP and SIGMA for edge–cloud fast–slow reasoning, using **simulation-based rollouts** to estimate the value of additional cloud reasoning and allocate queries to **decision-critical situations**.

AI for Science & AI for Optimization

- **LLM-Guided Black-Box Optimization** 2026 – Present *ICLR 2027, Under Review*
  Developed RULER, which uses LLMs to summarize solver trials into reusable **parameter-edit rules** and guide subsequent optimization through **hypothesis-driven experiments**.
- **Learning-Based Solver Selection Across Scales** 2026 – Present *ICLR 2027, Under Review*
  Developed a **size-agnostic solver policy** that adapts to shifting solver advantages during optimization, with theoretical sufficient conditions for a **runtime advantage** over fixed-solver strategies.

Physical World Intelligence

- **Wireless Foundation Models** Sept. 2024 – Mar. 2026 *IEEE Communications Magazine, Accepted*
  Developed WiLLM, a **unified foundation model** trained on **259K+ Wi-Fi sensing samples** from 12 datasets, enabling transferable representations across diverse devices, environments, and sensing tasks.

INTERNSHIP EXPERIENCE

AgiBot–AGIQUAD May 2026 – Oct. 2026
Embodied AI Research Intern; supervised by Qiwei Xu Shenzhen, China

- Developed WhichBench to diagnose VLN capability gaps in exact-instance grounding, informing a **data feedback loop** for targeted data collection and evaluation
- Built WhichGround, a **prototype embodied inspection agent** that orchestrates semantic-map/VLN systems as callable navigation skills, targeting future intelligent factory inspection agents
- Integrated the pipeline into the **D2 Max product platform** for **patrol and inspection scenarios**, demonstrating spoken-query and cross-floor navigation on a real quadruped. [Demo Video]

Shanghai AI Laboratory Jan. 2023 – Dec. 2024
Remote Research Intern; supervised by Xinyu Cai Remote

- Developed an **LLM-based agent pipeline** to generate OpenSCENARIO traffic scenarios from natural-language requirements; integrated CARLA with Apollo and prepared Waymo/V2X-ViT data
- Prepared **OCR training data for InternVL** by synthesizing challenging examples and filtering for quality; supported fine-tuning with part of the model frozen

Apple Inc. Jul. 2022 – Mar. 2023
Data Analyst Intern, Channel Planning and Forecast Beijing, China

- Built a modular Python/C++ allocation engine for weekly iPad distribution across authorized retail stores in China, **deployed in the iPad team's production planning workflow**
- Developed a two-stage forecasting pipeline using **geospatial clustering** followed by **random forest prediction** to support channel planning
- Automated allocation checks, AOS lookups, pricing, and launch reports for channel planning

PROJECT EXPERIENCE

6G Wireless Digital Twin and Simulation Sept. 2024 – Present
Project participant, Guangdong Major Project on Environment-Enhanced 6G Networks CUHK-Shenzhen

- Built a **Sionna RT** digital twin combining visual semantics, geometry, and measured data with **electromagnetic physics** to prioritize **high-fidelity simulation** under limited computing resources

Multi-Sensor UAV Docking and Adaptive Cruise Control System Oct. 2021 – Jan. 2023
Project Leader; supervised by Prof. Haiyuan Li

- Designed a distance-adaptive docking strategy using **GPS for long-range approach**, **vision for mid-range alignment**, and a **magnetic frame for close-range capture**
- Led integration and validation; won a university third prize and a Beijing "Internet+" third prize

TECHNICAL SKILLS

Programming: Python, C/C++, CUDA **Robotics & Simulation**: ROS, Isaac Sim
AI & Embodied Intelligence: PyTorch, RL, VLMs, WAMs, VLN, VLAs **Wireless Simulation**: Sionna RT

HONORS AND AWARDS

Outstanding Undergraduate Thesis, QMUL–BUPT Joint Programme Jun. 2024
Second Prize, BUPT Industrial Track, 8th China International "Internet+" Competition Jul. 2022
University-Level Third Prize; Beijing Regional "Internet+" Third Prize (UAV docking project)
First Prize, National Mathematics Competition for College Students Jul. 2021

ENGLISH PROFICIENCY

English: TOEFL iBT 103/120 (R27/L28/S23/W25), CET-6 516, CET-4 607

ACADEMIC SERVICE

Conference Reviewer: NeurIPS, ICML, ICLR, ICRA, IROS, AAAI, ACL/ARR, EMNLP, GLOBECOM, PIMRC.
Journal Reviewer: IEEE Transactions on Mobile Computing; IEEE Wireless Communications Letters.